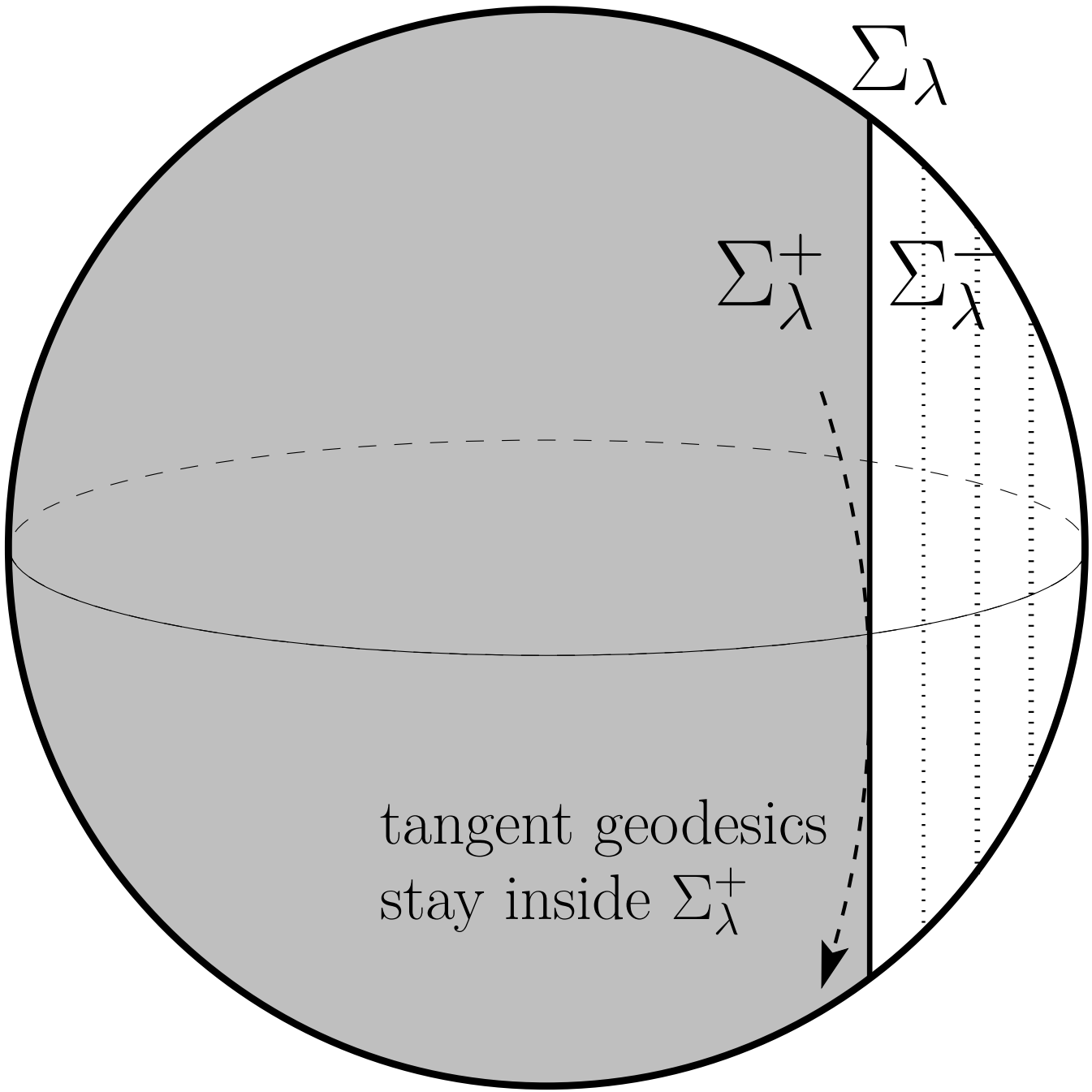

$\Sigma_\lambda$
$\Sigma_\lambda^+$
$\Sigma_\lambda^-$
tangent geodesics
stay inside $\Sigma_\lambda^+$